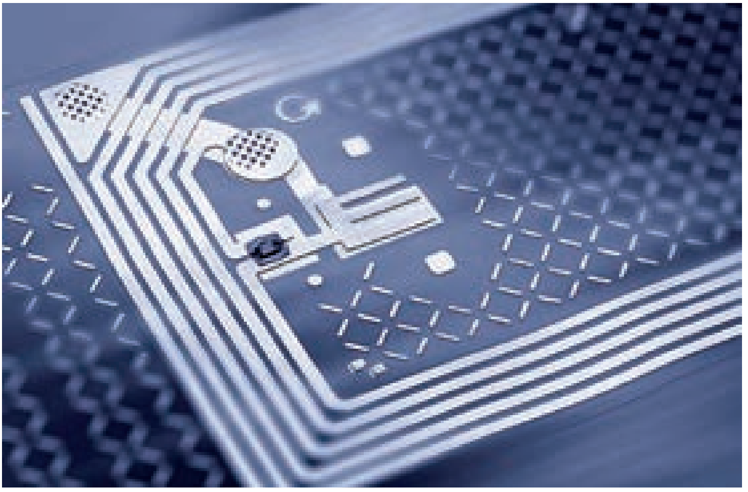

RFID microchip

2. When the tag's antenna receives electromagnetic energy from an RFID reader's antenna, the tag sends back radio waves to the reader.
3. These waves are decoded by the reader and data transfer is complete.

There are three types of RFID tags:

- **Active tags:** These RFID tags use internal batteries as a power source. They have a range of around 30 m which can be boosted to 100 m by using additional batteries.
- **Semi-passive tags:** They too have batteries but are only activated in the presence of a reader.
- **Passive tags:** Passive RFID tags rely entirely on the reader for power. They use the electromagnetic energy of the radio waves emitted by the reader. They are small in size and cheaper to manufacture. Because of the limited power source, they have a short range of 6 m.

RFID technology was originally meant to replace bar codes. This could potentially save thousands of man hours spent waiting in lines for check out. With RFID tags, you could just walk out of the store and the reader would automatically bill you for the products you've shopped.

It's currently being used to track livestock. This is done by fitting animals with location-tracking RFID chips. Pets are also being implanted with tiny RFID tags containing information about their owners and their medical history. More recently, these RFID tags are implanted in humans. These tags